My interest in healthcare AI started with a concern. After learning machine learning during my bachelor's degree, I became increasingly interested in what happens when models inherit the biases and limitations already present in medical research. I wrote about this in an article on gender bias in medical AI and reached out to researchers working in the field, proposing ways to evaluate fairness in clinical models. While I lacked the resources to pursue the work at the time, it shaped how I think about healthcare AI today. As models become increasingly powerful and increasingly opaque, understanding why a model reaches a conclusion is just as important as its accuracy from a safety perspective.

Satisfied with my ML knowledge base, I focused on applying it. I took a health innovations course covering the EU healthtech landscape and the end-to-end process of bringing innovations through it. A hospital project course investigating EHR interoperability challenges in Sweden, learning the gap between implemented and existing standards. A small consultancy project forecasting patient arrivals at an emergency department. Most recently, I've started a pilot with a hospital in Saudi Arabia on using AI to read electrophoresis results and speed up lab turnaround times. None are deep research projects individually, but the breadth of exposure means I understand the structures and common problems. A PhD coming from that awareness is more likely to produce something workers actually want to use, and that fits the system.

The technical work I'm most proud of in my master's is the multimodal pipeline I built for my thesis. I was classifying harmful comments on short-form video platforms, and a portion of the data was image and GIF-based. The pipeline ran a vision-language model to generate descriptions, then fed those into the text classifier. Because I was doing everything alone, from collecting, annotating, to building, I had a much better feel for what would work. I essentially thought, “What information helped me decide if something was harmful?”. So I came up with annotater notes, describing GIFs the same way I describe Gen Z slang to my parents, and it worked great for the accuracy. Similarly, I’d come across comments that, just by their content, could be harmless or harmful. But given the vibe of the entire comment section, it became clear what their intention is, so I included multiple passes in one of the architectures. I finally felt confident in making design choices and knew what was possible and how to implement it, compared to just running the latest models on the data.

The PhD position would require integrating pathology, MRI, genomics, biomarkers, and clinical information. A representation learning problem where each modality captures a different aspect of disease, often with conflicting signals, different levels of uncertainty and likely missing data. Building models that can reason across these modalities while remaining interpretable enough for clinical use is the correct way to build AI in healthcare.

Technically, I am strongest in machine learning, deep learning, and Python development. I have experience building transformer-based architectures, multimodal systems, and end-to-end machine learning pipelines, and I am comfortable working independently on open-ended

research problems. More importantly, I enjoy the process of exploring uncertain questions, identifying limitations in existing approaches, and designing methods to address them.

I would be excited to contribute to the Prostate Cancer Research Group and to develop methods that are not only technically strong but capable of supporting clinical decision-making in a meaningful and trustworthy way.